

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

A moving truck can tow a trailer if the combined weight of the trailer and the boxes it contains is no more than **4,600** pounds. What is the maximum number of boxes this truck can tow in a trailer with a weight of **500** pounds if each box weighs **120** pounds?

Answer

- A. **34**
- B. **35**
- C. **38**
- D. **39**

Correct Answer: A

Rationale

Choice A is correct. It’s given that the truck can tow a trailer if the combined weight of the trailer and the boxes it contains is no more than **4,600** pounds. If the trailer has a weight of **500** pounds and each box weighs **120** pounds, the expression **$500 + 120b$** , where **b** is the number of boxes, gives the combined weight of the trailer and the boxes. Since the combined weight must be no more than **4,600** pounds, the possible numbers of boxes the truck can tow are given by the inequality **$500 + 120b \leq 4,600$** . Subtracting **500** from both sides of this inequality yields **$120b \leq 4,100$** . Dividing both sides of this inequality by **120** yields **$b \leq \frac{205}{6}$** , or **b** is less than or equal to approximately **34.17**. Since the number of boxes, **b** , must be a whole number, the maximum number of boxes the truck can tow is the greatest whole number less than **34.17**, which is **34**.

Choice B is incorrect. Towing the trailer and **35** boxes would yield a combined weight of **4,700** pounds, which is greater than **4,600** pounds.

Choice C is incorrect. Towing the trailer and **38** boxes would yield a combined weight of **5,060** pounds, which is greater than **4,600** pounds.

Choice D is incorrect. Towing the trailer and **39** boxes would yield a combined weight of **5,180** pounds, which is greater than **4,600** pounds.